# Rakhi Singh

## Assistant Professor

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*Department of Mathematics*
*IIT Madras*
*Chennai, 600036, India*
*+91 44 2257 4610 (O)*
*rakhi@smail.iitm.ac.in*
*sites.google.com/view/singhrakhi*


## Education

2015–2018 **Ph.D. Statistics**, *IITB-Monash Research Academy*, CPI (coursework): 9.05/10
Advisors: Dr. Ashish Das (IITB, India) and Dr. Daniel Horsley (Monash University, Australia)

2008–2010 **M.Sc. Applied Statistics and Informatics**, *Indian Institute of Technology Bombay*
CPI: 8.37/10

2005–2008 **B.Sc. Mathematics, Statistics, and Computer Science**, *Banasthali University*
Percentage Marks: 83.0/100

## Experience

Jan.2025–present **Assistant Professor**, *IIT Madras, Chennai, India*

Aug.2022–Dec.2024 **Assistant Professor**, *Binghamton University, U.S.A.*

Jan.2020–Jul.2022 **Postdoc**, *University of North Carolina at Greensboro, U.S.A.*
Mentor: Dr. John Stufken

Jan.2019–Dec.2019 **Postdoc**, *Technical University of Dortmund, Dortmund, Germany*
Mentor: Dr. Joachim Kunert

July.2014–Dec.2014 **Research Assistant**, *Indian Institute of Technology Bombay, India*
Mentor: Dr. Ashish Das

Jul.2011–Jul.2014 **Business Analyst/Manager**, *American Express Pvt India Ltd, Gurgaon, India*

Jul.2010–Jul.2011 **Senior Programmer Analyst**, *EXL Services, Gurgaon, India*

## Research interests

**Subdata selection, screening experiments, choice experiments, fMRI designs, optimal design theory, rank aggregation problem, data science and interdisciplinary research**

## Publications

1. Zhang, F., Singh, R., and Stufken, J. (2026). An Extension of the GDS-ARM Algorithm for Factor Screening in Mixed-Level Supersaturated Designs. *J. Stat. Theory Pract.* 20, article 41. `https://doi.org/10.1007/s42519-026-00542-x`
2. Stallrich, J., Singh, R., Vogt-Lowell, K., and Li, F. (2025). Powerful Foldover Designs. *Qual. Reliab. Eng. Int.* 42 (2), 553 - 564. `https://doi.org/10.1002/qre.70105`
3. Singh, R. and Stufken, J. (2024). Factor selection in supersaturated designs by aggregation over random models. *Comput. Stat. Data Anal.* 194, 107940. `https://doi.org/10.1016/j.csda.2024.107940`
4. Singh, R. (2024). Pareto-efficient designs for multi- and mixed-level supersaturated designs. *Stat. Comput.* 34, article 38. `https://doi.org/10.1007/s11222-023-10354-9`

5. Singh, R. (2023). Best practices for multi- and mixed-level supersaturated designs. *J. Qual. Technol.* 56 (2), 113 - 127. https://doi.org/10.1080/00224065.2023.2259022
6. Singh, R. and Stufken, J. (2023). Subdata selection with a large number of variables. *The New England Journal of Statistics in Data Science* 1(3), 426 - 438. https://doi.org/10.51387/23-NEJSDS36
7. Singh, R. and Stufken, J. (2023). Selection of two-level supersaturated designs for main effects models. *Technometrics* 65, 96 - 104. https://doi.org/10.1080/00401706.2022.2102080
8. Chai, F.S., Singh, R., and Stufken, J. (2021). Connected row-column L-designs for symmetrical parallel line assays with two preparations. *J. Stat. Theory Pract.* 15, article 89. https://doi.org/10.1007/s42519-021-00219-7
9. Singh, R., Das, A., and Chai, F.S. (2021). On Three-Level A-Optimal Designs for Test-Control Discrete Choice Experiments. *Statistics and Applications* 19 (1), 199 - 208. https://ssca.org.in/media/15_19_1_2021_SA_Ashish_Das_3-level-revision.pdf
10. Singh, R., Kunert, J., and Stufken, J. (2021). On optimal fMRI designs for correlated errors. *J. Statist. Plann. Inference* 212, 84 - 96. https://doi.org/10.1016/j.jspi.2020.08.003
11. Singh, R. and Stufken, J. (2021). Efficient orthogonal fMRI designs in the presence of drift. *Stat. Methods Med. Res.* 30, 277 - 285. https://doi.org/10.1177/0962280220953870
12. Singh, R. and Kunert, J. (2021). Efficient crossover designs for non-regular settings. *Metrika* 84 (4), 497 - 510. https://doi.org/10.1007/s00184-020-00780-4
13. Singh, R., Dean, A., Das, A., and Sun, F. (2021). A-optimal designs under a linearized model for discrete choice experiments. *Metrika* 84 (4), 445 - 465. https://doi.org/10.1007/s00184-020-00771-5
14. Chai, F.-S., Das, A., Singh, R., and Stufken, J. (2020). Discriminating between superior UE($s^2$)-optimal supersaturated designs. *Statistics and Applications* 18 (2), 67 - 74. https://www.ssca.org.in/media/6_Vol._18_No._2_2020_Stufken.pdf
15. Singh R., Das, A., and Horsley, D. (2020). SUE($s^2$)-optimal supersaturated designs. *Statist. Probab. Lett.* 158, 108673. https://doi.org/10.1016/j.spl.2019.108673
16. Das, A. and Singh, R. (2020). A unified approach to discrete choice experiments. *J. Statist. Plann. Inference* 205, 193 - 202. https://doi.org/10.1016/j.jspi.2019.07.003
17. Chai, Feng-Shun, Singh, R., and Stufken, J. (2019). Nearly Magic Rectangles . *J. Combin. Des* 27 (9), 562 - 567. https://doi.org/10.1002/jcd.21667
18. Singh, R. and Mukhopadhyay, S. (2019). Exact Bayesian designs for count time series. *Comput. Stat. Data Anal.* 134, 157 - 170. https://doi.org/10.1016/j.csda.2018.12.008
19. Singh, R. (2019). On three-level D-optimal paired choice designs. *Statist. Probab. Lett.* 145, 127 - 132. https://doi.org/10.1016/j.spl.2018.09.005
20. Singh, R., Das, A., and Chai, F.S. (2019). Optimal Paired Choice Block Designs. *Stat. Sinica* 29, 1419 - 1438. https://doi.org/10.5705/ss.202016.0084
21. Das, A., Horsley, D., and Singh, R. (2018). Pseudo Generalized Youden Designs. *J. Combin. Des* 26 (9), 439 - 454. https://doi.org/10.1002/jcd.21594
22. Horsley, D. and Singh, R. (2018). New lower bounds for t-coverings. *J. Combin. Des* 26 (8), 369 - 386. https://doi.org/10.1002/jcd.21591
23. Cheng, C.S., Das, A, Singh, R., and Tsai, P.W. (2018). E($s^2$)- and UE($s^2$)-Optimal Supersaturated Designs. *J. Statist. Plann. Inference* 196, 105 - 114. https://doi.org/10.1016/j.jspi.2017.10.012
24. Chai, F.S., Das, A., and Singh, R. (2018). Optimal two-level choice designs for estimating main and specified two-factor interaction effects. *J. Stat. Theory Pract.* 12 (1), 82-92. https://doi.org/10.1080/15598608.2017.1329101

25. Dey, A., Singh, R., and Das, A. (2017). Efficient paired choice designs with fewer choice pairs. *Metrika* 80 (3), 309 - 317. `https://doi.org/10.1007/s00184-016-0605-9`
26. Chai, F.S., Das, A., and Singh, R. (2017). Three-level A- and D-optimal paired choice designs. *Statist. Probab. Lett.* 122, 211 - 217. `https://doi.org/10.1016/j.spl.2016.11.018`
27. Singh, R., Chai, F.S., and Das, A. (2015). Optimal two-level choice designs for any number of choice sets. *Biometrika* 102 (4), 967 - 973. `https://doi.org/10.1093/biomet/asv040`

## Softwares/Packages

1. Parker, H, Singh, R, Badal, PS. "RankAggSIgFUR: Polynomially Bounded Rank Aggregation under Kemeny's Axiomatic Approach." R Package version 1.0.0 (2022) `https://cran.r-project.org/web/packages/RankAggSIgFUR/index.html`
2. Singh, R, Stufken, J. "GDSARM: Gauss-Dantzig Selector - Aggregation over Random Models." R Package version 0.1.0 (2022) `https://cran.r-project.org/web/packages/GDSARM/index.html`

## Under revision/submitted/in preparation

1. Badal, P.S. and Singh, R. Heuristic Algorithms for Tied Kemeny Rank Aggregation (*submitted*).
2. Collins, D. and Singh, R. Subdata selection for high-dimensional big data (*submitted*).
3. Phillips, B., Singh, R. and Qiao, X. Subdata Selection for Principal Component Analysis (*submitted*).
4. Singh, R. and Stufken, J. Model-free tree-based subdata selection (*in preparation*).
5. Singh, R. and Badal, P.S. The R Package `RankAggSIgFUR` for Efficient Kemeny Rank Aggregation (*in preparation*).

## Courses taught

**IIT Madras**

**MA 5104: Statistical Inference and Learning**
teaching in Fall 2026

**MA 5755: Data Analysis & Visualization in R/Python/SQL**
taught in Fall 2026 – approx. student rating (0.90/1)

**MA 5750: Applied Statistics**
taught in Fall 2025 – approx. student rating (0.89/1)

**MA 5013: Applied Regression Analysis**
taught in Fall 2025 – approx. student rating (0.90/1)

**MA 2040/2103: Probability, Stochastic Processes and Statistics**
taught in Spring 2025 – approx. student rating (0.82/1)

**Binghamton University**

**MATH 455/531: Intro to Regression Models**
taught in Fall 2024, Spring 2023 – approx. student rating (4.40/5)

**MATH 556: Experimental Designs**
taught in Fall 2024, Fall 2023 – approx. student rating (4.93/5)

**MATH 532: Regression II**
taught in Spring 2024 – approx. student rating (4.49/5)

**MATH 448: Mathematical Statistics**
taught in Fall 2022 – approx. student rating (3.24/5)

**MATH 540: Capstone seminar**
taught in Fall 2024, Fall 2023

**UNC Greensboro**

**IAC 621: Data Science**
online development in Fall 2021; taught in Spring 2022

**IAA 621/STA 442/STA 642: Statistical Computing**
teaching in Fall 2021; online development in Spring 2021;
taught in Fall 2020 – approx. student rating (4.51/5)

**STA 432/STA 632: Introduction to Mathematical Statistics**
taught in Spring 2021 – approx. student rating (4.15/5)

**STA 431/STA 631: Introduction to Probability**
taught in Fall 2020 – approx. student rating (4.34/5)

**STA 352: Statistical Inference**
taught in Spring 2020 – approx. student rating (4.50/5)

**STA 290: Introduction to Probability and Statistical Inference**
taught in Spring 2020 – approx. student rating (4.14/5)

## Grants and awards

1. Bureau of Indian Standards, ₹9.96 lakh, for 2026 on "Review of Indian Standards of Basic and Advanced Statistical Design of Experiments with Analysis of variances".
2. Walmart Center for Tech Excellence (WCTE) Award, ₹20 lakh, for 2025-2026 on "Charting the Digital Pathway for Indian MSMEs".
3. University of Technology Sydney 2025 Key Technology Partnerships (UTS-KTP) Visiting Fellow Program Award, $A\$9,000$, for 2025-2026 on "Advancing Experiments to Reflect Time Perception in Health Valuation".
4. Binghamton University Data Science TAE Grant, $\$8,000$, for 2024-2025 (joint with Kenneth Kurtz) on "Advancing a statistical framework for the design of optimal experiments to evaluate formal psychological models of human category learning". Ended in Dec 2024 due to my move to IITM.
5. Binghamton University Data Science TAE Grant, $\$10,000$, for 2023-2024 (joint with Zimo Wang and Jia Deng) on "Advanced data analytic approach toward enabling autonomous nanofabrication platform".
6. AMS-Simons Travel Grant, $\$3,000$ per year, for 2023-2025 on "Supersaturated design constructions for screening experiments". Ended in Dec 2024 due to my move to IITM.
7. New Faculty Grant, UNC Greensboro, $\$5,000$, for 2021–2022 (joint with J.Stufken) on 'Information-Based Optimal Subdata Selection for Big Data With Large Numbers of Variables'.
8. Excellence in Ph.D. Research Award 2019 by IIT Bombay, India.
9. Best 3-Minute Thesis Award 2015 by IITB-Monash Research Academy, India.
10. Best Poster Award 2015 by IITB-Monash Research Academy, India.
11. Young Scientist Award 2015 by Society of Statistics, Computer and Applications (SSCA), India.
12. Chairman's Award For Innovation 2014 by American Express Pvt India Ltd, Gurgaon, India.
13. Analyst of the Quarter Q1 2012 and Q2 2013 by American Express Pvt India Ltd, Gurgaon, India.

## Invited talks/contributed talks/posters at conferences/universities

1. Invited (did not attend): DAE, Rutgers University, USA, 2026.
2. Invited: Dr. B. R. Ambedkar National Institute of Technology Jalandhar, India, 2026.
3. Seminar: (Chief Guest on National Mathematics Day) Hindustan Institute of Technology & Science, India, 2025.
4. Seminar: Indian Institute of Technology Madras, India, 2025.
5. Invited: University of Technology Sydney, Australia, 2025.

6. Invited (did not attend): AISC, Greensboro, USA, 2024.
7. Invited: DAE, Virginia Tech, USA, 2024.
8. Invited: Joint Research Conference, U of Waterloo, Canada, 2024.
9. Invited: CM Statistics, Berlin, Germany, 2023.
10. Invited: Fall Technical Conference (FTC), Raleigh, USA, 2023.
11. Invited: Quality and Productivity Research Conference (QPRC), Houston, USA, 2023.
12. Invited: Spring Research Conference (SRC), Banff, Canada, 2023.
13. Invited: International Conference on Design of Experiments (ICODOE), Memphis, USA, 2023.
14. Invited: EcoSta, Japan, virtual, 2022.
15. Seminar: Simon Fraser University, Canada, Jan 2022.
16. Seminar: Binghamton University, USA, Jan 2022.
17. Seminar: University of Texas at San Antonio, USA, Jan 2022.
18. Seminar: Miami University, USA, Jan 2022.
19. Seminar: Rochester Institute of Technology, USA, Jan 2022.
20. Seminar: University of Waterloo, Canada, Jan 2022.
21. Seminar: University of Illinois at Chicago, USA, Jan 2022.
22. Seminar: Wake Forest University, USA, Dec 2021.
23. Seminar: University of Texas at El Paso, USA, Dec 2021.
24. Invited: International Conference on Advances in Interdisciplinary Statistics and Combinatorics, virtual, 2021.
25. Invited: Design and Analysis of Experiments (DAE), virtual, 2021.
26. Invited: LinStat, virtual, 2021.
27. Invited: Workshop on Quality Improvement Methods (Farewell celebration of Joachim Kunert), virtual, 2021.
28. Invited: International Indian Statistical Association Conference (IISA), virtual, 2021
29. Invited: CM Statistics, virtual, 2020.
30. Seminar: Otto von Guericke University (2019), Magdeburg, Germany, 2019.
31. Invited: Model-oriented data analysis and optimum design (mODa), Smolenice, Slovakia, 2019.
32. Invited: International Conference on Design of Experiments (ICODOE), Memphis, USA, 2019.
33. Contributed: International Conference on Risk Analysis and Design of Experiments, Vienna, Austria, 2019.
34. Invited: Conference on Experimental Design and Analysis (CEDA), Hsinchu, Taiwan, 2018.
35. Seminar: Arizona state University, Tempe, USA, 2018.
36. Invited: International Conference on Advances in Interdisciplinary Statistics and Combinatorics, Greensboro, USA, 2018.
37. Contributed: Society of Statistics, Computer and Applications, Puducherry, India, 2018.
38. Contributed: Society of Statistics, Computer and Applications, Jammu, India, 2017.
39. Contributed: Society of Statistics, Computer and Applications, Jammu, India, 2016.
40. Contributed: Australasian Conference on Combinatorial Mathematics and Combinatorial Computing, Newcastle, Australia, 2016.
41. Contributed: International Indian Statistical Association Conference (IISA), India, 2015.
42. Poster: Designed Experiments: Recent Advances in Methods and Applications (DEMA), Sydney, Australia, 2015.
43. Contributed: Australasian Conference on Combinatorial Mathematics and Combinatorial Computing, Brisbane, Australia, 2015.

44. Poster: Design and Analysis of Experiments (DAE), Raleigh, USA, 2015.
45. Invited: Society of Statistics, Computer and Applications, Bhubaneshwar, India, 2015.

## Student advising

### Ph.D. students

**Anant Shinge**, *2026–present, at IITM*
Advanced topics in screening designs

**Naveen Kumar S**, *2026–present, at IITM*
Constructions and analysis of screening designs

**Vinay Kumar Sharma**, *with S. Mukhopadhyay, 2026–present, at IITB*
Spatial Sampling Designs

**Pramit Majumder**, *with P.S. Badal, 2025–present, at IITM*
Multi-Hazard Assessment Framework for Indian Metropolises

**David Collins**, *with X. Qiao, 2023–present, at Binghamton University*
Subdata selection for high-dimensional and multivariate data

**Bruce Phillips**, *with X. Qiao, 2023–present, at Binghamton University*
Advanced topics in subdata selection

### Masters students

**Aritra Dasgupta**, *Aug.2026–Apr.2027, at IITM*, M.Tech. project
Design-based explainable AI

**Amar Kumar**, *Aug.2025–Apr.2026, at IITM*, M.Tech. project
Building Faster Oblique Random Forest Package in R

**Santanu Das**, *Aug.2025–Apr.2026, at IITM*, M.Tech. project
Hybrid Quantum Machine Learning for Time Series Forecasting Anomaly Detection

**Hannah Parker**, *with P.S. Badal, Jan.2022–Apr.2022, at UNCG*, Capstone Advisor
Exploring the state-of-art in the Rank Aggregation problem

### Undergraduate students

**M. Aaryan Sriram**, *with P.S. Badal, Jul.2026–present, at IITM*, IITM Young Research Fellow (YRF)
Hyperparameter optimization for deep neural networks and random forests

**Shayan Sareen**, *Jul.2025–Jun.2026, at IITM*, IITM YRF
Hyperparameter optimization for deep neural networks and random forests

**Atharva Moghe**, *with P.S. Badal, Jul.2025–Jun.2026, at IITM*, IITM YRF
Developing Efficient Algorithms for Aggregating Rankings and Finding the Top-k Objects

**Ethan Yattaw**, *Jan.2024–Apr.2024, at BU*, BU Projects for New UG Researchers
Design-Based Hyperparameter Optimization for Machine Learning Models

### Other students

**Afroze Nazira**, *Mar.2025–Sep.2025, at IITM*, Project Associate
Charting the digital pathway for Indian MSMEs

**Anushka Mathur**, *Mar.2025–present, at IITM*, Project Associate
Charting the digital pathway for Indian MSMEs

**Rhys O'Higgins and Samuel Griffin**, *with J.Stufken, Summer 2022, at UNCG*, REU
Subdata selection

**Samantha Kilcoyne and Trisha Nayak**, *with J.Stufken, Summer 2021, at UNCG*, REU
Data compression and prediction

## Service

### At IIT Madras

2025. Core committee member for the new online program by the Department of Mathematics.

2025. Curriculum committee member for the Department of Mathematics.

### At Binghamton University

2022. Panelist for 'Honoring Women in STEM panel', an event conducted by *Association For Women in Mathematics* at BU

2022–2024. Statistics seminar organizer

2023–2024. Computers Committee

2022–2024. Statistics Committee, Actuarial Science Committee

2023–2024. Faculty advisor for Digital and Data Studies minor in Harpur College

2023–2024. Lecturer Search Committee for Digital and Data Studies minor in Harpur College

2023–2024. Computing & Network Committee, Undergraduate Advising Committee

2024. Graduate Committee, Master of Statistics Admission Committee

### At UNC Greensboro

2021. Served on the 'Staff Search Committee' for the MS in Informatics and Analytics

2021–2022. Served on the 'Admissions Committee' for the MS in Informatics and Analytics

### To the profession

2026. Organizing an invited session at the Design and Analysis of Experiments meeting.

2024. Organized one invited session at Joint Statistical Meetings (JSM) 2024.

2024–present. Associate Editor for the J. Statistical Theory and Practice.

2023–present. Associate Editor for the J. Agricultural, Biological, and Environmental Statistics.

2023. Organized and chaired one DAE session at CM Statistics 2023.

2023. Chaired one DAE session at QPRC 2023.

2023. Chaired one DAE session at ICODOE 2023.

2022. Organized four and chaired one DAE sessions at AISC 2022 (organized jointly with A. Mandal and J. Stufken).

2022. Organized and chaired a topic-contributed session (Subsampling: Basic tool that facilitates the identification of statistical relationships in big data) at the JSM 2022.

2021. Panel moderator for the panel on the future of design and analysis of experiments for the conference 'Design and Analysis of Experiments (DAE) 2021'

2021. Website developer for the conference 'Design and Analysis of Experiments (DAE) 2021'.

2021. Served on the local organizing committee at the International Conference on Advances in Interdisciplinary Statistics and Combinatorics 2021, virtual.

2021. Organized and chaired an invited paper session (Screening important variables in sparse setups while considering interactions) at the 63rd ISI World Statistics Congress 2021, virtual.

2020–2021. Guest Editor (with A. Mandal and J. Stufken) for a special issue on design of experiments in the *Journal of Statistical Theory and Practice*

2019–present. Reviewer for several journals including Journal of American Statistical Association (2), Statistica Sinica (7), Journal of Royal Statistics Society C (1) and B (1), Technometrics (1), Statistics and Computing (1), Journal of Computational and Graphical Statistics (1), Sankhya (1), Metrika (2), Journal of Statistical Planning and Inference (3), Journal of Applied Statistics (2), Journal of Combinatorial Designs (3), Statistics (2), Statistical Papers (3), Journal of Statistical Theory and Practice (3), Communications in Statistics - Simulation and Computation (1) etc.